

Splunk Investigation

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Findings

Time: 2025-10-15 12:45:00 UTC – 2025-10-15 13:15:00 UTC

Host: FRONTDESK-PC1

Compromised Account: Ryan.Adams

Indicators of Compromise:

- IOC IPs: 172[.]16[.]0[.]184 (Source), 157[.]245[.]46[.]190 (C2)
- Location of file: C:\Users\Ryan.Adams\Music\python.exe
- Filename: python.exe
- Persistence mechanism (if any): Scheduled Task

Investigation

On 2025-10-15 13:00:00 UTC, the user Ryan Adams reported that their mouse was moving independently from their own mouse movement, indicating that the account may have been compromised. Approximately 12:51:00 UTC, it was discovered that multiple accounts had failed to log on to the FRONTDESK-PC1 host, indicating a brute force attack was attempted. At 12:55:17 UTC it was also discovered that the Ryan.Adams account was successfully logged onto by the source IP 172[.]16[.]0[.]184.

Upon the login, it was also discovered that the account had administrative privileges. At 12:51:27 UTC, the threat actor (TA) began to change the Windows defender configuration settings, specifically real-time protection was disabled at 12:55:00 UTC. The threat actor then began to retrieve, and execute a malicious file known as **python.exe** in Ryan Adam's music folder. Once the file was executed, the file opened a remote connection to the TA's C2 server (IP: 157[.]245[.]46[.]190), which confirms the user's complaints of mouse movement. Furthermore, it was also detected at 13:04:59 UTC that the TA used PowerShell to create a scheduled task known as **PythonUpdate** that will persistently run the RAT upon startup. Additionally, no further evidence of compromise was detected.

WHO –

- Host: FRONTDESK-PC1
- Attacker IP: 172[.]16[.]0[.]184, with 157[.]245[.]46[.]190 as the C2

WHAT – Domain accounts were password sprayed, and the Ryan.Adams account was successfully logged into by a remote host.

WHEN. At 12:51:17 UTC the TA gained initial access to the Ryan.Adams account, and modified Windows Defender Real-time Protection at 12:55:00 UTC. The attacker also created a scheduled task at 13:04:59 UTC following the execution of python.exe

WHERE: The activity took place on the host: FRONTDESK-PC1, specifically the "C:\\Users\\Ryan.Adams\\Music" directory of the Ryan.Adams user.

WHY: The intent of the TA maybe to gain domain access and maintain persistence. The TA could also intend to gain access to the DC. Additionally, the Ryan.Adams account may have had weak credentials.

HOW:

- The Attacker performed a password spray against many accounts
- The Ryan.Adams account was successfully compromised by the attacker
- The Attacker logged into the account
- The TA modified Windows Defender configuration settings
- The attacker retrieved and execution the python.exe file
- The attacker utilized PowerShell to maintain persistence by creating a scheduled task.

Recommendations

1) Immediately Isolate the host machine and remove the python.exe file and remove the scheduled task.

2) Begin to reset the account credentials for the Ryan.Adams account. Run a query searching for the IPs and file downloaded to identify other hosts exhibiting similar behaviors. If additional hosts were found, immediately isolate them as well.

3) Although IP addresses are relatively easy to change for an attacker, consider placing these domains in a blocklist to prevent additional compromise.